

Exchange Rate Regimes in Production Networks

A Small Open Economy Extension of Rubbo (2024)

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Building on: Rubbo (2024) *Networks, Phillips Curves, and Monetary Policy*

Outline

1. Intro

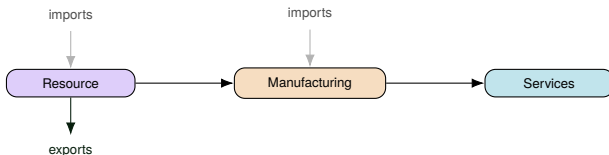
2. Model

3. Results

4. Conclusion

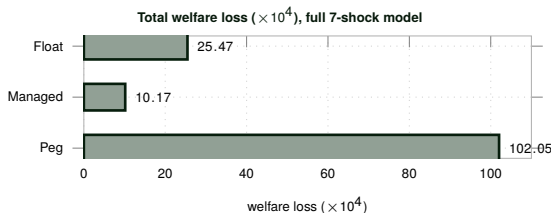
Motivation

- Network Phillips curves (Rubbo 2024, La'O–Tahbaz-Salehi 2022): **closed economy**, unique DC index
- Open-economy NK (Galí–Monacelli 2005): FX channel, **no network**
- **This paper: both together** — multi-sector network + FX



Questions: Optimal FX regime? Does the DC index survive a 2nd cost-push channel?

Headline Result



- **Managed float dominates** both corners
- **Peg dominated:** zero autonomy, UIP shock passes straight through
- **Driver:** persistence (near-unit-root NFA), not volatility

Units: consumption-equivalent loss, $\times 10^{-4}$. Calibrated to Chile's real national input-output table (Banco Central de Chile, Cuadros 12x12, CdeR 2018, ref. year 2023) — full calibration and sources on the next two slides. Solved as the genuine nonlinear model (Dynare), not linearized.

Literature

Network side (closed economy):

- **Rubbo (2024)** — Calvo network $\Rightarrow$ unique DC index (1 cost-push channel)
- **La'O & Tahbaz-Salehi (2022)** — optimal MP in production networks
- **Pasten, Schoenle & Weber (2020)** — sectoral rigidity heterogeneity drives MP transmission

Open-economy side:

- **Galí & Monacelli (2005)** — 1-sector SOE: domestic-inflation targeting $\approx$ optimal, peg dominated (ToT channel)
- **Fanelli & Straub (2021)** — FX intervention as a *second* instrument, separate from i_t
- **Schmitt-Grohé & Uribe (2003)** — debt-elastic premium to stationarize NFA; flags 1st-order welfare risk
- **Gopinath & Itskhoki (2022); Amiti et al. (2019); Auer, Levchenko & Sauré (2019)** — pass-through, dominant currency, IO spillovers
- **Itskhoki & Mukhin (2023); Bianchi & Coulibaly (2025); Coulibaly (2023)** — frontier: managed float/fear-of-floating *optimal*, not behavioral (no network)
- **Gnecato, Montes-Galdón & Stamato (2025, ECB); Kalemli-Özcan et al. (2025)** — open economy *with* network (tariffs), fixed Taylor rule, no FX-regime choice¹
- **Qiu, Wang, Xu & Zanetti (2026, JME)** — closest paper: derives the open-economy DC index directly extending Rubbo, WIOD 43-country calibration; but *static* (no NFA/persistence), no UIP, and no FX-regime choice — their own conclusion lists relaxing financial autarky as their top open extension

Gap: network + open economy + FX **regime choice** together — no one has this. Same ranking as Galí–Monacelli but a **different channel** (risk-premium, not ToT); Managed’s dominance echoes Fanelli–Straub & Itskhoki–Mukhin, but from network persistence.

→ paper-by-paper comparison to the literature: Appendix

¹ Details & what these papers cite for open-economy production networks: Appendix.

This Paper

- Extend Rubbo → SOE: imported inputs Ω^F , UIP, debt-elastic premium
- Ask: optimal FX regime? DC index survive with 2 cost-push channels?
- **Three literatures, none overlapping fully** (network only; open economy only; network + open economy but no regime choice) — **first to put all three together**
- **Not just Gali-Monacelli with sectors**: ranking survives, but the mechanism (risk-premium) and sector heterogeneity are new
- Calibrate: Chile's real I-O table (Banco Central), also Korea, Czechia & a stylized baseline; solve **nonlinear** model directly

Model: Overview

Households

- CRRA consumption, disutility of labor; **Euler eq.:** intertemporal consumption smoothing; labor supply: wage = MRS
- **UIP** (derived, not assumed): domestic rate = foreign rate + risk premium + expected depreciation

Firms

- Cobb-Douglas: labor + domestic network inputs + imports
- **Marginal cost** moves with A_{it} (TFP) and S_t (import cost) — **two cost-push channels**
- **Calvo pricing:** reset prob. δ_i , forward-looking reset price; sector-specific stickiness → heterogeneous slopes

Open Economy

- **Law of one price:** import price = $S_t \times$ foreign price; **UIP** closes the FX block
- **NFA law of motion:** current account accumulates foreign bonds; exports: foreign demand $\times$ relative-price elasticity

Key takeaway: same network structure as Rubbo (2024), **plus** an exchange-rate cost-push channel running through imported inputs.

→ full equations & notation: Appendix

Model: Policy Regimes and Shocks

Regime	Rule	Exchange rate
Float	$i_t = \bar{i} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$	fully endogenous (UIP)
Peg	$S_t = \bar{S}$	fixed; i_t residual
Managed	$i_t = \bar{i} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \log(S_t/\bar{S})$	partially stabilized

Shocks (AR(1)): a_{it} (TFP, $\rho=0.90$) p_t^{F*} (import price, $\rho=0.85$) D_t^* (foreign demand, $\rho=0.80$)

Theory: Regime Comparison — Predictions

	Float	Peg	Managed
Exchange rate	endogenous	fixed	partial
FX cost-push absorbed by	S_t	π_i, y	split
Monetary independence	full	none	full

Naive: float best. **Overtuned** — see Welfare section.

Does the DC Index Survive a Second Cost-Push Channel?

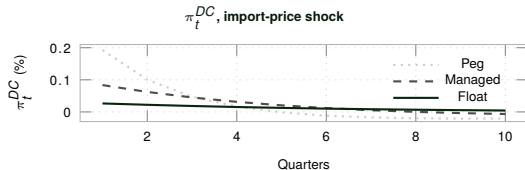
Closed economy (Rubbo): 1 channel (TFP) $\Rightarrow$ unique DC index

$$\phi^\top \nu = \mathbf{0}^\top \Rightarrow \pi_t^{DC} = \sum_i w_i^{DC} \pi_{it}$$

Open economy: 2 channels (TFP + FX) $\Rightarrow$ need *both*, same weights ϕ

$$\underbrace{\phi^\top \nu = \mathbf{0}^\top}_{\text{TFP}} \text{ and } \underbrace{\phi^\top \Gamma = 0}_{\text{FX}} \Rightarrow \text{generically infeasible}$$

- **Answer: no**, generically
- This calibration: $\cos(\Gamma, \mathcal{B}) \approx 0.96$ — close, but not constant
- **Fix:** flexible import pricing ($\hat{\delta}_M=1$) restores π_t^{DC}



Full DC-index algebra, proof, and knife-edge test: Appendix.

Calibration: Chile, Real IO Data

Parameter	Value	Parameter	Value
β	0.99	$\alpha_1, \alpha_2, \alpha_3$	0.503, 0.359, 0.604
γ, φ	1.00, 2.00	$\Omega_1^F, \Omega_2^F, \Omega_3^F$	0.077, 0.195, 0.070
η	1.50	$\beta_1^H, \beta_2^H, \beta_3^H$	0.027, 0.229, 0.744
ε	8.00	$\delta_1, \delta_2, \delta_3$	0.90, 0.31, 0.16
θ^*	2.00	ψ	0.020
$\beta^{F,tot}$	0.10	$\phi_\pi, \phi_\gamma, \phi_s$	1.50, 0.50, 0.30

Dense Ω^H (order: Res., Man., Serv.), all 9 entries used, no triangular assumption:

$$\Omega^H = \begin{pmatrix} 0.075 & 0.153 & 0.193 \\ 0.099 & 0.202 & 0.145 \\ 0.002 & 0.058 & 0.266 \end{pmatrix}, \quad \text{Export share} = (0.602, 0.180, 0.036)$$

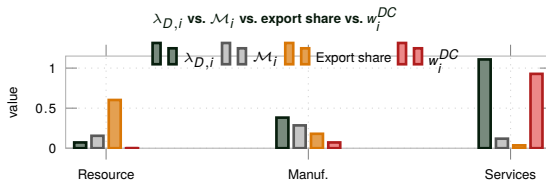
- Resource = mining/copper (Chile's export engine, 60% of own output exported) → Manuf. (mid-chain) → Services (largest value added, sticky)²
- $\alpha_i, \Omega_{ij}^H, \Omega_i^F, \beta_i^H$: **data** (Banco Central de Chile, Cuadros 12x12, CdeR 2018, ref. 2023)
- $\delta_1, \delta_2, \delta_3$: **literature** — euro-area price-change frequencies (Dhyne et al. 2005/ECB IPN) → quarterly reset prob.; cross-checked vs. Nakamura & Steinsson (2008) US PPI
- $\beta, \gamma, \varphi, \eta, \varepsilon, \theta^*, \psi, \phi_\pi, \phi_\gamma, \phi_s$: literature-standard. Solved nonlinear, order 1

² Sector concordance (Banco Central de Chile, CdeR 2018, 12 activities → our 3): **Resource** = {1 Agriculture, forestry & fishing, 2 Mining}; **Manufacturing** = {3 Manufacturing industry, 4 Electricity, gas, water & waste management, 5 Construction}; **Services** = {6 Trade, hotels & restaurants, 7 Transport, communications & information services, 8 Financial intermediation, 9 Real estate & housing services, 10 Business services, 11 Personal services, 12 Public administration}. Import & export data exist at the sector level from the same tables: imports = imported intermediate input use per buying sector (sheet 21, underlies Ω_i^F); exports = exports of each sector's own output (sheet 20, underlies the export-share figures above).

Calibration: Network Properties

$$\lambda_{D,i} = [(\beta^H)^\top (I - \Omega^H)^{-1}]_i, \quad \mathcal{M}_i = [(I - \Omega^H)^{-1} \Omega^F \mathbf{1}]_i, \quad w_i^{DC} \propto \lambda_{D,i} \frac{1 - \delta_i}{\delta_i}$$

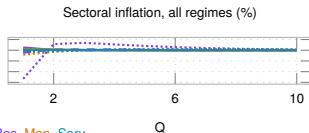
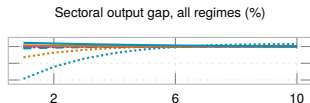
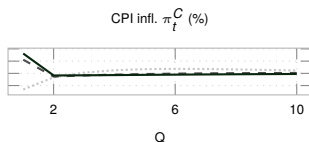
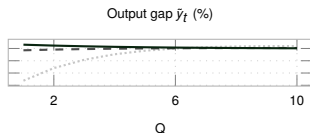
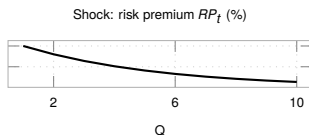
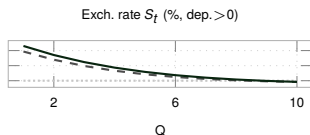
$\lambda_{D,i}$: Domar weight $\mathcal{M}_i$: import centrality w_i^{DC} : DC weight



- FX exposure runs **opposite** to Domar weight
- Resource: small weight (0.07), most FX-exposed ($\mathcal{M}_i = 0.16$) — and *is* the export engine (60% of own output exported, copper)
- Services: dominates output ($\lambda_D = 1.11$), FX-exposed only **indirectly**; w_i^{DC} is essentially a Services index (0.93)

IRFs: Foreign Risk-Premium Shock — the Key Mechanism

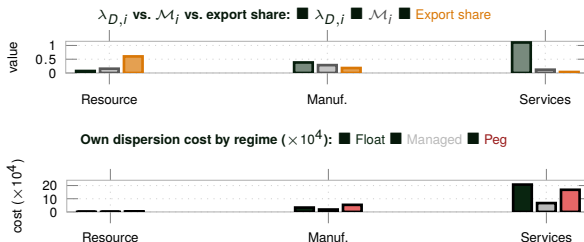
The shock that decides the ranking: 75% of Peg's welfare loss (Chile calibration).



Peg ... Managed - - Float — Sectoral by regime: line style = regime, color = Res. Man. Serv.

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Network Exposure & Sector Welfare Preferences

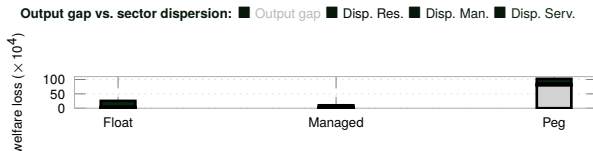
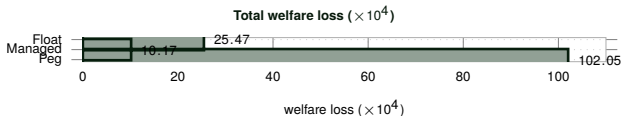


Exposure $\rightarrow$ pass-through $\Gamma_i = \frac{[\Delta(I - \Omega\Delta)^{-1}\omega_F]_i}{\kappa_i}$, welfare weight $\kappa_i = \lambda_{D,i} \in \frac{1 - \delta_i}{\delta_i}$:

	$\mathcal{M}_i$	Export share	Γ_i	κ_i
Resource	0.155	0.602	1.091	0.07
Manuf.	0.285	0.180	0.001	21.5
Services	0.119	0.036	<0.0001	276.3

- Under literature-sourced Calvo stickiness ($\hat{\delta}_3 = 0.03$), κ_i concentrates in Services — FX pass-through barely matters for Services' welfare cost *relative to its own weight*: almost all of Γ_i funnels through Resource
- Managed dominates every sector.** Peg costliest for Resource (0.37) & Manuf. (5.36); for Services, **Float is costliest** (20.73 > Peg's 16.79) — near-unit-root S_t under Float feeds the stickiest sector's marginal cost for a long time

Welfare: Headline & Persistence

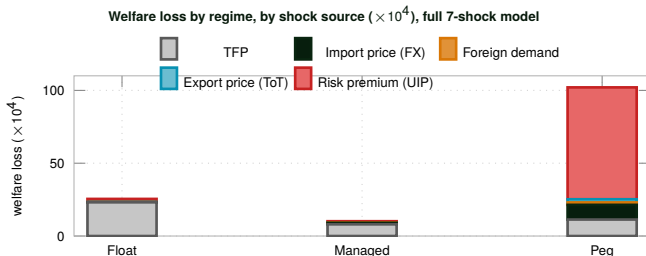


$\mathcal{W} = \frac{1}{2} \left[(\gamma + \varphi) \text{Var}(\tilde{y}_t) + \sum_i \lambda_{D,i} \frac{1 - \delta_i}{\delta_i} \text{Var}(\pi_{it}) \right]$ — **Peg dominated** on output gap & dispersion, every sector.

Why: persistence, not volatility.

- Float's S_t (near-unit-root NFA) feeds *every* sector's MC, even for domestic shocks
- Peg's output-gap term explodes (79.5) because it has zero monetary autonomy to absorb the risk-premium shock
- Slow decay, not larger impact, drives Float's Services dispersion above Managed's

Shock Decomposition of Welfare Cost



- **Risk-premium/UIP dominates Peg** (75% of its total)
- **Ranking:** Managed < Float $\ll$ Peg — Peg no longer competitive

2nd-order check: this is the channel a genuine order-2 solution could move most — it does move most, but only +0.7% (see Conclusion & appendix).

Mechanisms: Why Each Regime Costs What It Costs

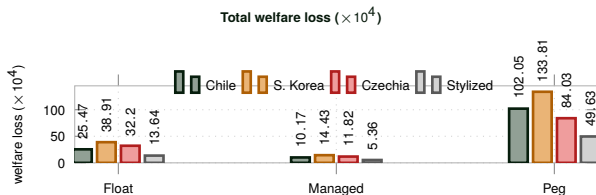
Regime	Cost channel	What sets the magnitude
Float	S_t 's near-unit-root persistence (NFA state) feeds $\Gamma_i \Delta e_t$ into <i>every</i> sector's MC, even for shocks with no FX content — pure dispersion cost	Persistence of debt-elastic premium (ψ); Γ_i (import exposure); κ_i of the most central/sticky sector (Services: 20.7 of 25.5)
Peg	Zero monetary autonomy: output-gap term explodes (79.5 of 102.1, $\approx 78\%$) on the UIP shock; $\Delta e_t \equiv 0$ <i>relocates</i> , not removes, adjustment into upstream P^H	Absence of ϕ_π, ϕ_y ; real vs. nominal share of the shock; upstream sector's Domar weight
Managed	$\phi_S \Delta \log S_t$ damps $\Gamma_i \Delta e_t$ without fully re-imposing peg's relocation cost — dominates every sector at once	Interior optimal $\phi_S^* \approx 0.20$: too low $\rightarrow$ UIP shock leaks in; too high $\rightarrow$ autonomy lost

Two forces, always in tension: Γ -channel favors peg (switches off $\Gamma_i \Delta e_t$); **relocation** channel favors float (peg's fixed e_t forces the same adjustment into upstream prices). Which wins depends on Γ_i vs. κ_i ; managed is the only regime improving on *both* corners for every sector.

Analytical or simulated? Both.

- **Channels — closed-form:** $\Gamma_i = [\tilde{A}\omega_F]_i / \kappa$ (pass-through), $\kappa_i = \lambda_{D,i} \varepsilon \frac{1-\delta_i}{\delta_i}$ (welfare weight), DC-survival $\phi^\top \mathcal{V} = 0^\top$ and $\phi^\top \Gamma = 0$ — from primitives alone, no model solve.
- **Magnitudes/ranking — need simulation:** $\text{Var}(\pi_{it}), \text{Var}(\tilde{y}_t)$ come from the full linear RE solution (order-1 Dynare) — no closed form once $\phi_S > 0$ or Ω^H dense. Managed $\ll$ Float $\ll$ Peg and $\phi_S^* \approx 0.20$ are simulated, but robust across all four calibrations.

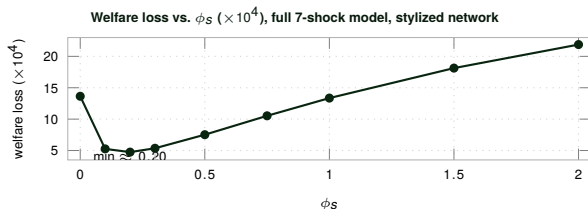
Robustness: South Korea, Czechia & the Stylized Baseline



- **South Korea:** diversified manufacturing exporter (Bank of Korea, 2022), deeper manufacturing import intensity than Chile ($\Omega_{\text{manuf}}^F = 0.24$ vs. 0.19)
- **Czechia:** third real IO calibration (Eurostat, 2022), deepest manufacturing import intensity ($\Omega_{\text{manuf}}^F = 0.32$, EU/German supply chains)
- **Stylized:** original hand-picked triangular Ω^H (Rubbo-style, no data) — checks the result isn't an artifact of network shape
- **Ranking survives in all four:** Managed < Float $\ll$ Peg

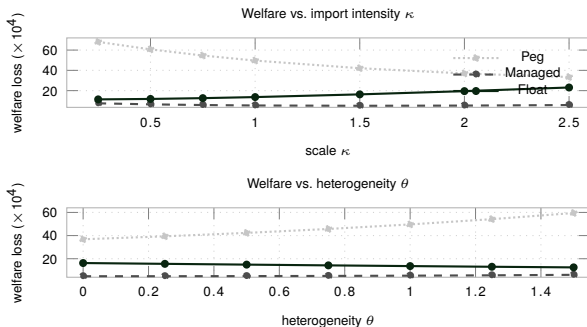
UIP dominates Peg's loss in all four: 75% Chile, 72% Korea, 65% Czechia, 81% stylized — not specific to one calibration. All solved nonlinear (Dynare), not linearized.

Generalization: Optimal FX Stabilization ϕ_S



- **Why interior:** too low $\rightarrow$ UIP shock leaks in; too high $\rightarrow$ loses autonomy
- $\phi_S=0 \rightarrow 0.20$: loss 13.64 $\rightarrow$ 4.74. **Peaked, not flat:** 0.20 beats calibrated 0.30 by $\approx 13\%$
- Beyond ≈ 0.2 : rises fast, back above Float by $\phi_S=1$

Generalization: Robustness



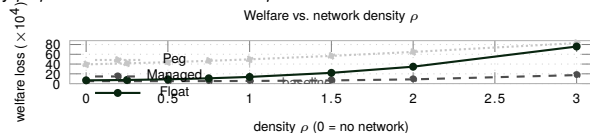
Managed < **Float** < **Peg** robust (stylized network, as in the FX-stabilization sweep), but the gap moves in **opposite directions**:

- κ (import scale): Peg falls, Float rises — **converge**
- θ (concentration): Peg rises, Float falls — **diverge**

Isolating the Network Channel

New sweep, stylized triangular network (not Chile's dense Ω^H): scale domestic I-O links

OH_{21}, OH_{32} by ρ . $\rho=0$: no domestic network. $\rho=1$: baseline.



Network welfare premium by sector ($\times 10^4$), total($\rho=1$)—total($\rho=0$): ■ Resource ■ Manuf. ■ Services



- **Managed is almost network-neutral:** $\rho=0 \rightarrow 1$ adds only +0.35 to Managed vs. +6.98 Float, +10.30 Peg
- **Denser network $\Rightarrow$ corners diverge from Managed:** by $\rho=3$, Float (75.9) catches Peg (82.9), both $\gg$ Managed (17.4)
- **Burden lands on different sectors:** Peg/Managed hit upstream Resource/Manuf., Services gains; Float hits Services most
- **Verdict:** network **triples** the regime gap as density rises ($\rho:1 \rightarrow 3$) and **redirects** who bears the cost

Conclusion

- **Managed float optimal:** 10.17 vs. Float 25.47, Peg 102.05 ($\times 10^4$, Chile calibration)
- **Driver:** risk-premium/UIP dominates Peg (75%); TFP persistence dominates Float & Managed
- **Optimal** $\phi_s \approx 0.20$: sharply peaked — over-stabilizing S_t is costly
- **Ranking robust** to κ, θ, ρ (network density) & to Korea/Czechia calibrations; the *gap* is not
- **Network channel isolated** (stylized network, ρ dial): $\rho=0 \rightarrow 1$ adds only +0.35 to Managed vs. +6.98 Float, +10.30 Peg — **denser networks favor Managed most**, and reallocate the network's sectoral burden (Services gains under Peg, loses under Float)
- **Vs. literature:** same ranking as Galí–Monacelli, different channel (risk-premium, not ToT)
- **2nd-order-verified** (Dynare order=2, pruned, simulated): loss rises $\leq 2\%$ net of MC noise, ranking unchanged — see appendix
- **Next:** many-sector OECD TiVA calibration (beyond the 3-sector Chile/Korea/Czechia data)

Appendix: Is the Ranking Robust to a Genuine 2nd-Order Solution?

Why this was a caveat: Rubbo's LQ welfare formula is 2nd-order in utility, but evaluated on *1st-order* (certainty-equivalent) paths — valid only if $\mathbb{E}[\tilde{y}_t] = \mathbb{E}[\pi_{it}] = 0$ exactly. Two open-economy features can break that: the **debt-elastic UIP wedge** (nonlinear in B_t^*) and the **near-unit-root NFA** process — both mean the true *risk-adjusted* steady state can sit away from the deterministic one.

Method: the model is already coded nonlinear (levels) in Dynare. Solve to **order 2 with pruning, simulate** 260k periods (analytic order- ≥ 2 moments are unavailable: the Taylor rule leaves price levels/S with a unit root). Compute $\mathbb{E}[X^2]$ directly (not Var) so any risk-adjusted mean shift is captured. Control for simulation noise with an identical-seed order=1 run.

Welfare loss ($\times 10^4$)	Float	Managed	Peg
1st-order (analytic, headline)	25.47	10.17	102.05
1st-order (simulated, same seed)	25.58	10.20	102.39
2nd-order (pruned, simulated)	26.09	10.26	103.13
Net 2nd-order effect	+2.0%	+0.6%	+0.7%

- **Ranking survives:** Managed < Float < Peg at 2nd order, correction $\leq 2\%$ everywhere

Appendix: DC Index — Definition and Insulation

Closed economy (Rubbo): 1 channel (TFP), $\mathcal{V}$ rank-deficient $\Rightarrow$ unique DC index:

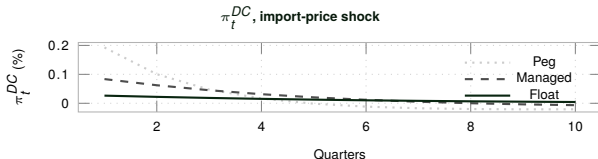
$$\phi^\top \mathcal{V} = \mathbf{0}^\top \Rightarrow \pi_t^{DC} = \sum_i w_i^{DC} \pi_{it}, \quad w_i^{DC} \propto \lambda_{D,i} \frac{1-\delta_i}{\delta_i}$$

Open economy: 2 channels (TFP + FX):

$$\underbrace{\phi^\top \mathcal{V} = \mathbf{0}^\top}_{\text{TFP}} \text{ and } \underbrace{\phi^\top \Gamma = 0}_{\text{FX}} \Rightarrow \text{generically infeasible}$$

Knife-edge test: $\Gamma \propto \mathcal{B}$ would save the DC index. Instead: $\cos(\Gamma, \mathcal{B}) \approx 0.96$ ($\approx 17^\circ$ apart) — close, but $\Gamma_i/\mathcal{B}_i$ shrinks $\approx 2\times$ per sector, not constant. Breakdown is real, not a numerical artifact.

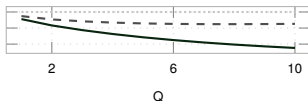
Fix: import nodes flexible ($\delta_M=1 \Rightarrow w_M=0$) $\Rightarrow$ DC well-defined again by construction — what the Taylor rules target throughout (π_t^{DC}); FX insulation itself is a quantitative question, shown below.



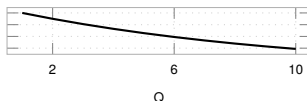
Float: smallest on impact (0.026%) & most persistent; Peg/Managed overshoot and reverse sign — on-impact insulation $\neq$ full story.

IRFs: Resource TFP Shock

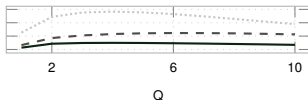
Exch. rate S_t (% , dep. > 0)



Shock: TFP A_{1t} (%)



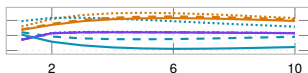
Output gap $\tilde{y}_t$ (%)



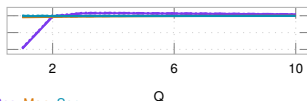
CPI infl. π_t^C (%)



Sectoral output gap, all regimes (%)



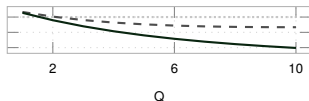
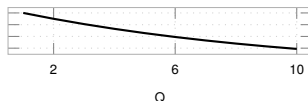
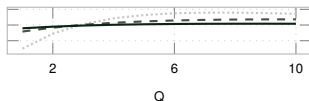
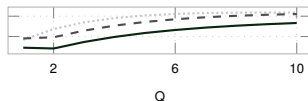
Sectoral inflation, all regimes (%)



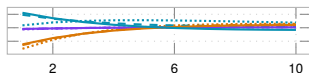
Peg ... Managed - - Float — Sectoral by regime: line style = regime, color = Res. Man. Serv.

Q

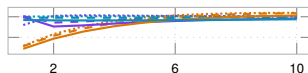
IRFs: Manufacturing TFP Shock

Exch. rate S_t (% , dep. > 0)Shock: TFP A_{2t} (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

Sectoral output gap, all regimes (%)



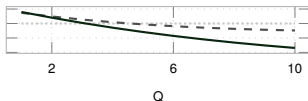
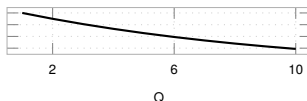
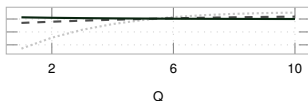
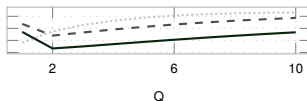
Sectoral inflation, all regimes (%)



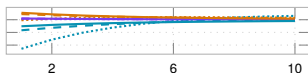
Peg ... Managed - - Float — Sectoral by regime: line style = regime, color = Res. Man. Serv.

Q

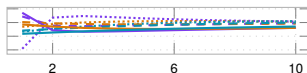
IRFs: Services TFP Shock

Exch. rate S_t (% , dep. > 0)Shock: TFP A_{3t} (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

Sectoral output gap, all regimes (%)



Sectoral inflation, all regimes (%)

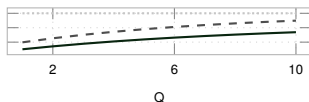


Peg ... Managed - - Float — Sectoral by regime: line style = regime, color = Res. Man. Serv.

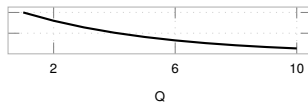
Q

IRFs: Foreign-Demand Shock

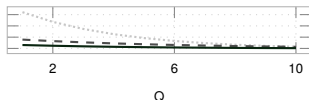
Exch. rate S_t (% , dep. > 0)



Shock: foreign demand D_t^* (%)



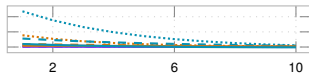
Output gap $\tilde{y}_t$ (%)



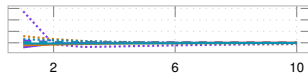
CPI infl. π_t^C (%)



Sectoral output gap, all regimes (%)



Sectoral inflation, all regimes (%)

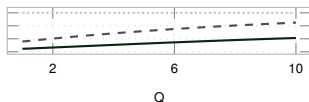


Peg ... Managed - - Float — Sectoral by regime: line style = regime, color = Res. Man. Serv.

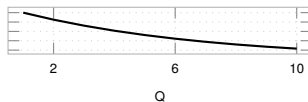
Q

IRFs: Export-Price / ToT Shock

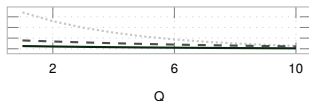
Exch. rate S_t (% , dep. > 0)



Shock: export price P_t^X (%)



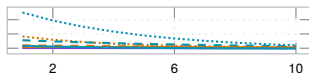
Output gap $\tilde{y}_t$ (%)



CPI infl. π_t^C (%)



Sectoral output gap, all regimes (%)



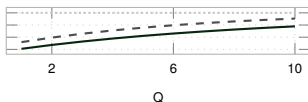
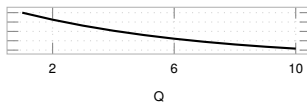
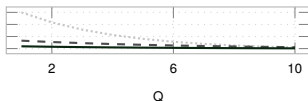
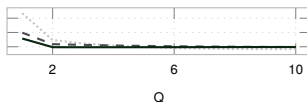
Sectoral inflation, all regimes (%)



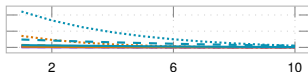
Peg ... Managed - - Float — Q Sectoral by regime: line style = regime, color = Res. Man. Serv.

Q

IRFs: Import-Price Shock (Backup)

Exch. rate S_t (% , dep. > 0)Shock: import price P_t^{F*} (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

Sectoral output gap, all regimes (%)



Sectoral inflation, all regimes (%)



Peg ... Managed - - Float — Q Sectoral by regime: line style = regime, color = Res. Man. Serv.

Q

Appendix: Model Equations & Notation (1/2)

Households

$$\max \mathbb{E}_0 \sum_t \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right] \quad \text{s.t.} \quad P_t^C C_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i^*) \left[1 - \psi(\hat{b}_{t-1}^* - \bar{b}^*) \right] S_t B_{t-1}^*$$

$$C_t^{-\gamma} = \beta(1+i_t) \mathbb{E}_t \left[C_{t+1}^{-\gamma} / \Pi_{t+1}^C \right], \quad W_t / P_t^C = C_t^\gamma L_t^\varphi, \quad (1+i_t) = (1+i^*) \left[1 - \psi(\hat{b}_t^* - \bar{b}^*) \right] \mathbb{E}_t[S_{t+1}/S_t] \text{ (UIP)}$$

C , L : consumption, labor

β , γ , φ : discount, RRA, inv. Frisch

W , i : wage, dom. rate

S , i^* : ER, foreign rate

ψ : debt-elastic premium

B^* : foreign bond

P^C : CPI

Π_t : profits (rebated)

Firms: Technology and Cost

$$Y_{it} = A_{it} L_{it}^{\alpha_i} \prod_j X_{ijt}^{\Omega_{ij}^H} M_{it}^{\Omega_i^F} \Rightarrow MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

$$W_t L_{it} = \alpha_i MC_{it} Y_{it}, \quad P_{jt} X_{ijt} = \Omega_{ij}^H MC_{it} Y_{it}, \quad S_t P_t^{F*} M_{it} = \Omega_i^F MC_{it} Y_{it}$$

Y , A : output, TFP

L , X , M : labor, dom./imp. inputs

Ω_{ij}^H : dom. IO share

Ω_i^F : import cost share

α_i : labor share

MC_{it} : nominal marg. cost

P^{F*} : foreign import price

Key: MC_{it} moves with both A_{it} (TFP) and S_t (via M_{it}) — the two cost-push channels.

Appendix: Model Equations & Notation (2/2)

Firms: Calvo Pricing — reset price (probability δ_i resets, $1 - \delta_i$ keeps last price):

$$X1_{it} = \widetilde{MC}_{it} Y_{it} + (1 - \delta_i) \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \Pi_{i,t+1}^\varepsilon X1_{i,t+1}, \quad X2_{it} = Y_{it} + (1 - \delta_i) \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \Pi_{i,t+1}^{\varepsilon-1} X2_{i,t+1}$$

$$P_{it}^\# / P_{it} = \frac{\varepsilon}{\varepsilon - 1} X1_{it} / X2_{it}, \quad 1 = (1 - \delta_i) \Pi_{it}^{\varepsilon-1} + \delta_i \left(P_{it}^\# / P_{it} \right)^{1-\varepsilon}, \quad \delta_i = \frac{\delta_i (1 - \beta(1 - \delta_i))}{1 - \beta \delta_i (1 - \delta_i)}$$

δ_i : Calvo reset prob. (primitive)

δ_i : derived slope (enters DC weights)

$P_{it}^\#$: reset price

ε : elasticity (markup $\varepsilon / (\varepsilon - 1)$)

Open Economy

$$\underbrace{P_t^{FH} = S_t P_t^{F*}}_{\text{LOP}}, \quad \underbrace{(1+i_t) = (1+i^*) \left[1 - \psi(\hat{b}_t^* - \bar{b}^*) \right] \mathbb{E}_t[S_{t+1}/S_t]}_{\text{UIP}}$$

$$\hat{b}_t^* = \frac{1+i_t-1}{\Pi_t^C} \hat{b}_{t-1}^* + \frac{P_t^H EX_t - P_t^{FH} IM_t}{P_t^C}, \quad EX_t = \kappa_{EX} D_t^* \left(\frac{P_t^H}{P_t^{FH}} \right)^{-\theta^*}$$

$$Y_{it} = \beta_i^H \frac{P_t^H}{P_{it}} \left(C_t^H + EX_t \right) + \sum_j \Omega_{ji}^H \frac{MC_{it} Y_{jt}}{P_{it}}$$

$\hat{b}^*$: NFA (scaled)

EX, IM : exports, imports

θ^* : export elasticity

D^* : foreign demand shifter

β_i^H : cons. share, sector i

Float: i_t Taylor rule, S_t free

Peg: $S_t = \bar{S}$, i_t residual

Appendix: The Peg vs. Float Debate — What Do We Know?

Why it depends (structure):

- **Mundell (1961)** OCA — fix favored by small, open, factor-mobile economies w/ correlated shocks vs. anchor; float favored by large, diversified ones
- **Frankel (1999)** — “no single regime is right for all countries or at all times”; openness, size, trade & capital-market structure all matter

Shock-absorption view (why float wins on real shocks — robustly):

- **Friedman (1953)** — S_t adjusts instead of costly wage/price deflation
- **Broda (2004)** — 75 dev. countries, 1973–96: negative ToT shocks → large real GDP declines under pegs
- **Edwards & Levy-Yeyati (2005)** — 10% ToT deterioration: −0.80pp growth under pegs vs. −0.43pp under floats
- **Céspedes, Chang & Velasco (2004)** — even with dollarized-liability balance-sheet effects, float *still* beats fixed

But floating is not a free lunch (limits on float's insulation):

- **Calvo & Reinhart (2002)** “Fear of Floating” — EMEs peg de facto anyway: credibility-building, not literal welfare optimum
- **Rey (2013)** “Dilemma not trilemma” — under free capital mobility, float alone doesn't deliver monetary autonomy

Consensus: no corner solution is universally optimal — the choice trades a nominal anchor against real-shock absorption, shock- and structure-dependent.

Appendix: The Frontier — Is Fear of Floating Actually Optimal?

- Itskhoki & Mukhin (2023)** — general open-economy framework with endogenous UIP/PPP deviations: efficient policy = MP closes the output gap + FX intervention eliminates the UIP wedge. Optimal policy is generically a **managed float/crawling peg**, matching observed fear of floating. When the natural RER is stable, this collapses to an “open-economy divine coincidence.”
 ↪ vs. *us*: **same headline** (Managed, not a corner, is optimal) — reached instead via a multi-sector Calvo network with a persistence-driven, not UIP-gap-closing, mechanism
- Bianchi & Coulibaly (2025; 2022 WP)** — fear of floating is **optimal**, not behavioral, when credit access is tied to the real exchange rate (self-fulfilling crisis risk). Prescription: float for real shocks, **fix** for financial shocks
 ↪ vs. *us*: **opposite ranking** for our financial shock — our risk-premium/UIP shock passes straight through under **Peg** (worst regime). Different financial friction (persistent NFA wedge vs. collateral-constraint crisis) flips the prescription
- Coulibaly (2023)** — discretionary MP is procyclical to offset balance-sheet effects; commitment (inflation targeting) or capital controls raise welfare
 ↪ vs. *us*: echoes our “Managed = extra instrument” logic — giving policy more tools beats a corner regime

Bottom line: the 2023–25 frontier has also moved past corner solutions — but *why* fear of floating is optimal is mechanism-specific. We add production-network amplification + shock persistence as a distinct reason, not examined by either paper.

Appendix: How Our Results Compare to the Literature

Literature	Says	Us
Mundell (1961); Frankel (1999)	Optimal regime depends on structure, no universal answer	Confirms: ranking flips with shock type (UIP vs. TFP) and network density
Friedman (1953); Broda (2004); Edwards–Levy–Yeyati (2005); Céspedes–Chang–Velasco (2004) Galí & Monacelli (2005)	Float buffers real (ToT/foreign) shocks, lower output loss than peg — robust even to balance-sheet effects Peg dominated, float $\approx$ optimal in 1-sector SOE	Same direction (Peg worst), different shock: risk-premium/UIP, not ToT Same ranking of Peg , but pure Float is not optimal: Managed beats it
Calvo–Reinhart (2002)	EMEs avoid pure float — read as fear , a credibility-driven deviation from optimum	We find partial stabilization is literally welfare-optimal — a persistence-driven, not fear-driven, reason
Rey (2013)	Float alone doesn't insulate; need a second instrument (capital controls/macropuru)	Echoes Fanelli–Straub: Managed uses S_t itself as that second instrument
Qiu, Wang, Xu & Zanetti (2026, JME)	Open-economy DC index (CPI+expenditure-switching+profit channels), <i>static</i> , WIOD-calibrated; DC breaks down, output-gap targeting near-optimal	Independent corroboration DC breaks down in open economy; we add the piece they flag as their own top open extension — UIP, NFA persistence, and FX-regime choice

Appendix: The ECB's Open-Economy Production-Network Model

Gnocoato, Montes-Galdón & Stamato (2025), “Tariffs across the supply chain,” ECB WP No. 3081 — also underlies Lane’s (ECB Exec. Board) 13 May 2026 speech on energy shocks (4-region extension):

- **Setup:** two-region, multi-sector, full IO structure, international production network; sticky wages/prices; Taylor rule on *domestic* CPI (no regime comparison)
- **Finding:** tariffs on **intermediate** inputs → persistent inflation & larger GDP loss (low substitutability, network propagation); tariffs on **final** goods → one-off inflation spike, smaller GDP loss — *where* in the network a shock hits sets its persistence
- ⇔ *vs. us:* their tariff-location result and our TFP-vs-UIP result are the same insight in two settings: shock **location/type** sets its persistence, echoing our “driver is persistence, not volatility”

What it cites for open economy + production networks:

- Network amplification, closed economy: **Pasten, Schoenle & Weber (2020)**; **Ghassibe (2021)**; **Rubbo (2023)**³; **Afrouzi & Bhattarai (2023)**
- Open-economy multi-sector NK + networks: **Devereux, Gente & Yu (2023)**; **Hinterlang et al. (2023)** EMuSe (Bundesbank); closest: **Kalemli-Özcan, Soylu & Yildirim (2025)** “Global networks, MP and trade”
- Trade fragmentation & inflation: **Moro & Nispi Landi (2024)**; **Ambrosino, Chan & Tenreiro (2026, JIE)**
- Optimal MP under tariffs: **Bergin & Corsetti (2023)**; **Bianchi & Coulibaly (2025)** “Optimal MP response to tariffs” (*different* paper from their fear-of-floating one)

Bottom line: none compares FX **regimes** or asks whether a DC index survives — they fix the monetary rule. ~~That gap is still exactly our question.~~

³ Published version: Rubbo, E. (2023), *Econometrica* 91(4), 1417–1455 — journal year is 2023, not 2024.